

Train the Model for Spam / Ham Emails

Spam Email Detector App

Enter or Paste the text of Email

Subject: 🎉 Congratulations! You've Won a \$1,000 Gift Card!

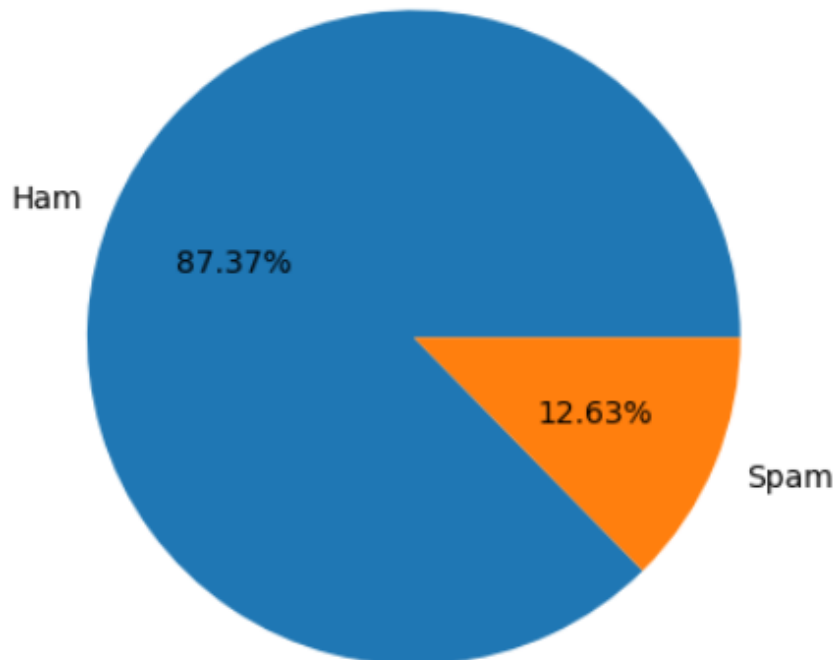
From: rewards@claim-prize-now.com

Predict

This is a Spam Email

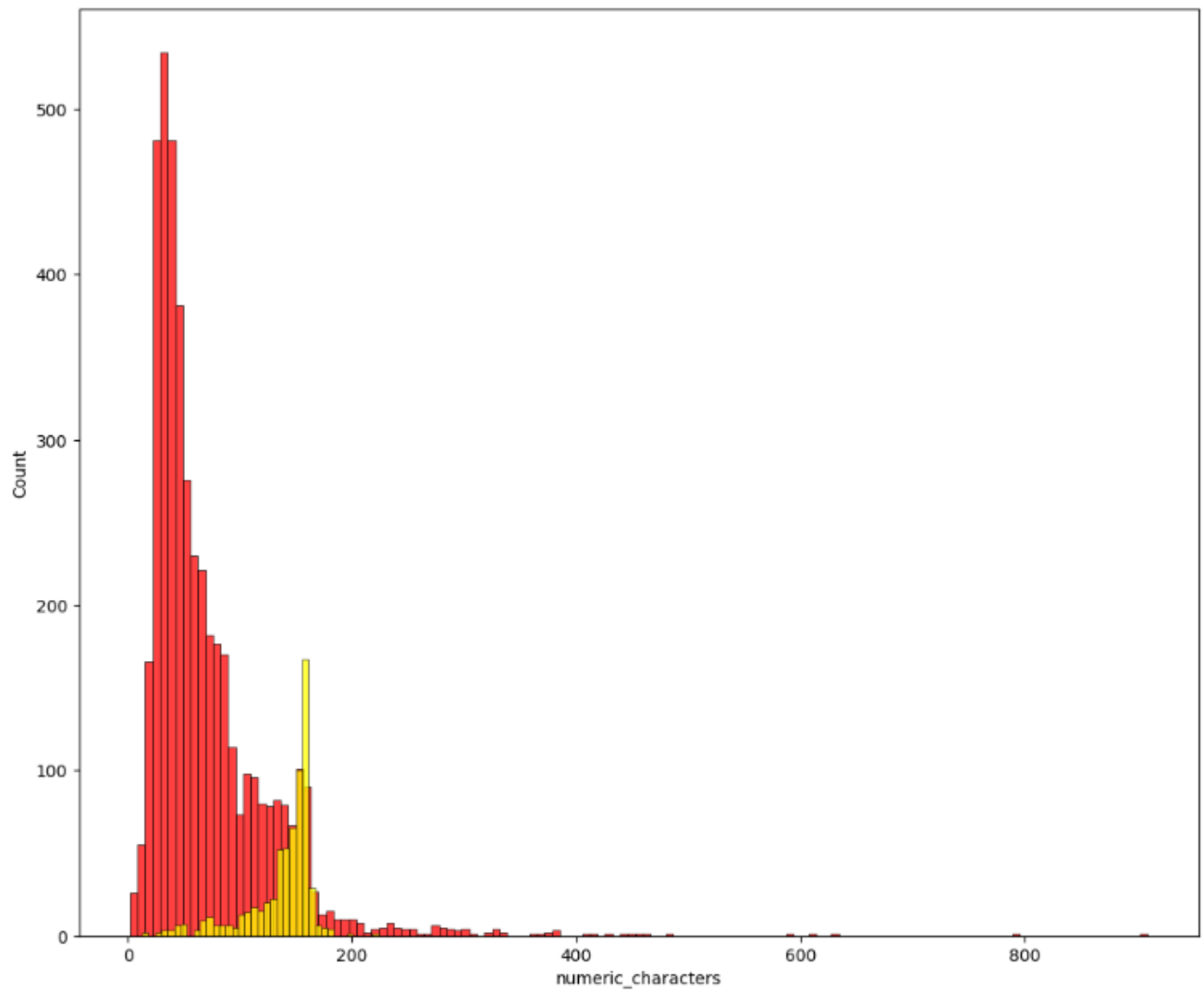
Draw a pie chart which represent % of ham & spam

Data Is Imbalance Because Ham Is 87.37% And Spam 12.63%



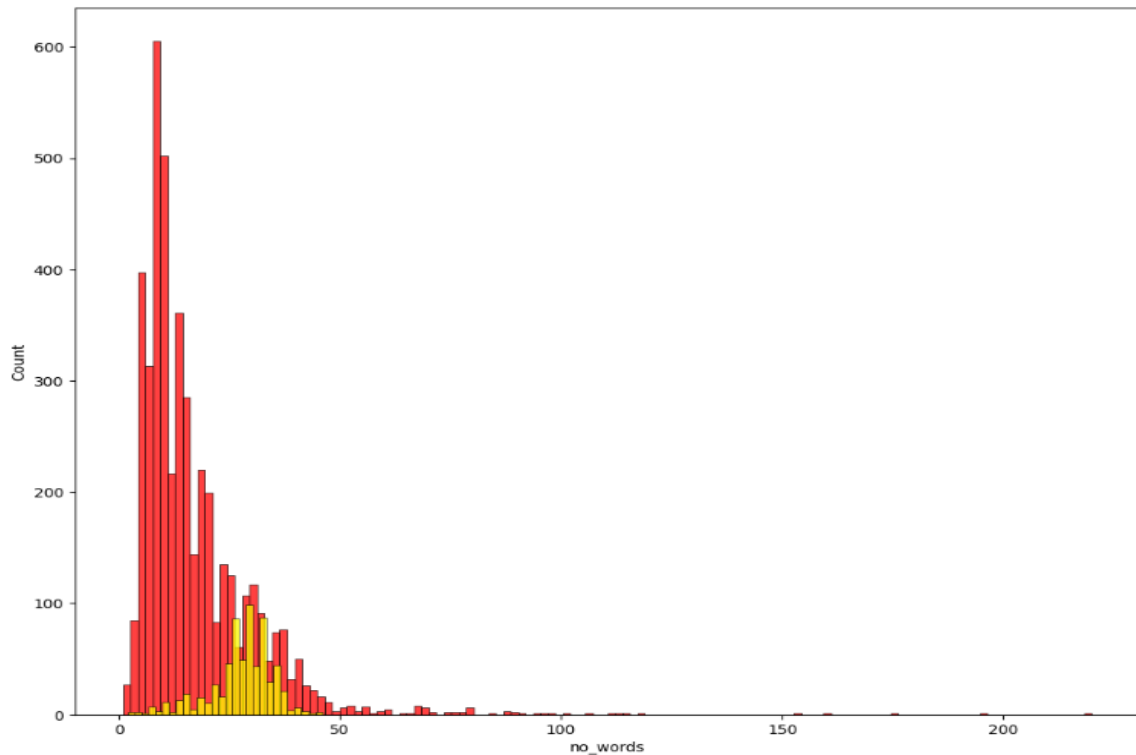
Analysis By Histogram :: No Of Characters

No Of Characters In Spam Messages More Than No Of Characeters In Ham Messages



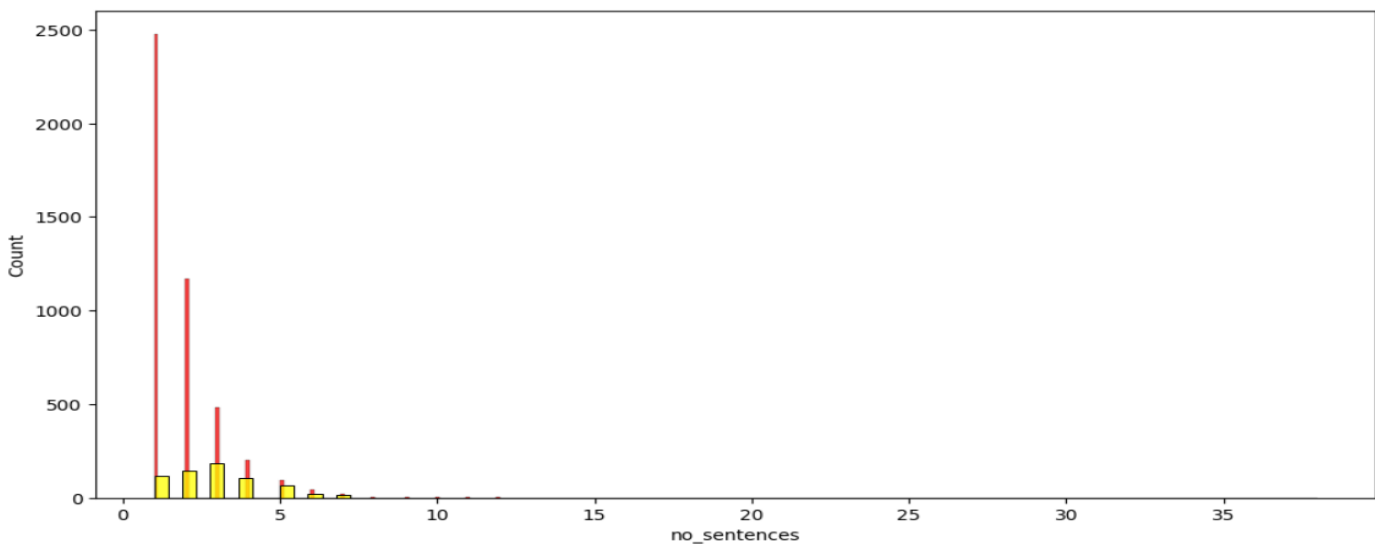
Analysis by Histogram :: no of words

No Of Words In Spam Messages More Than No Of Words In Ham Messages



Analysis by Histogram :: no of sentences

No Of Sentences In Spam Messages More Than No Of Sentences In Ham Messages

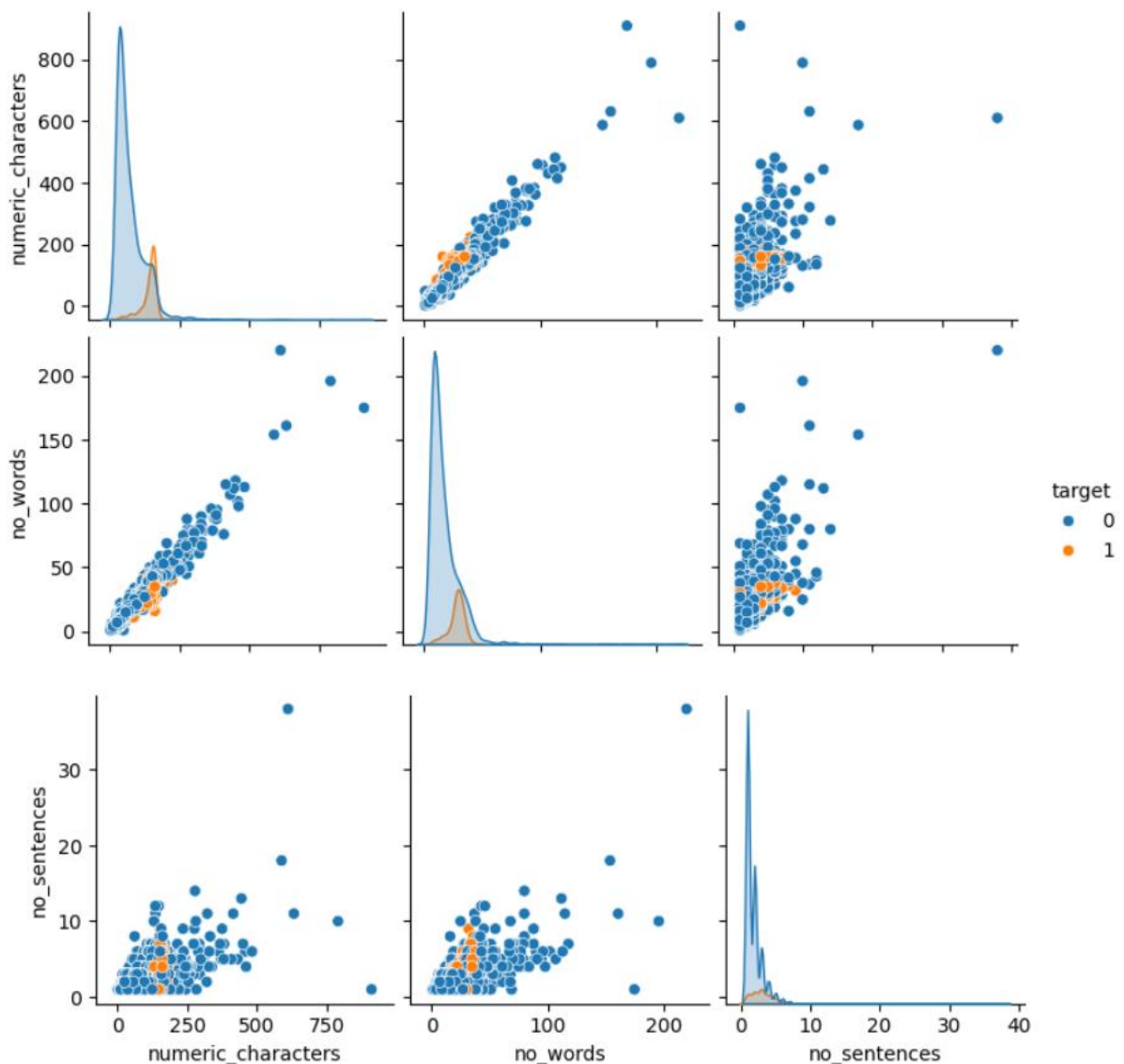


Pair Plot : Relationship Between 3 Columns No_Word, Numeric_Characters, No_Sentences By Pair Graph

Strong +Ve Correlation Between Numeric_Characters And No_Words Means - Messages Contain More Words And More Characters

Spam Messages Are Clustered At Higher Values Of Numeric_Characters, No_Words, And No_Sentences

There Are Some Messages Overlap Between Spam and Ham , But Spam Messages Are More Concentrated In The Higher Ranges Of These Features



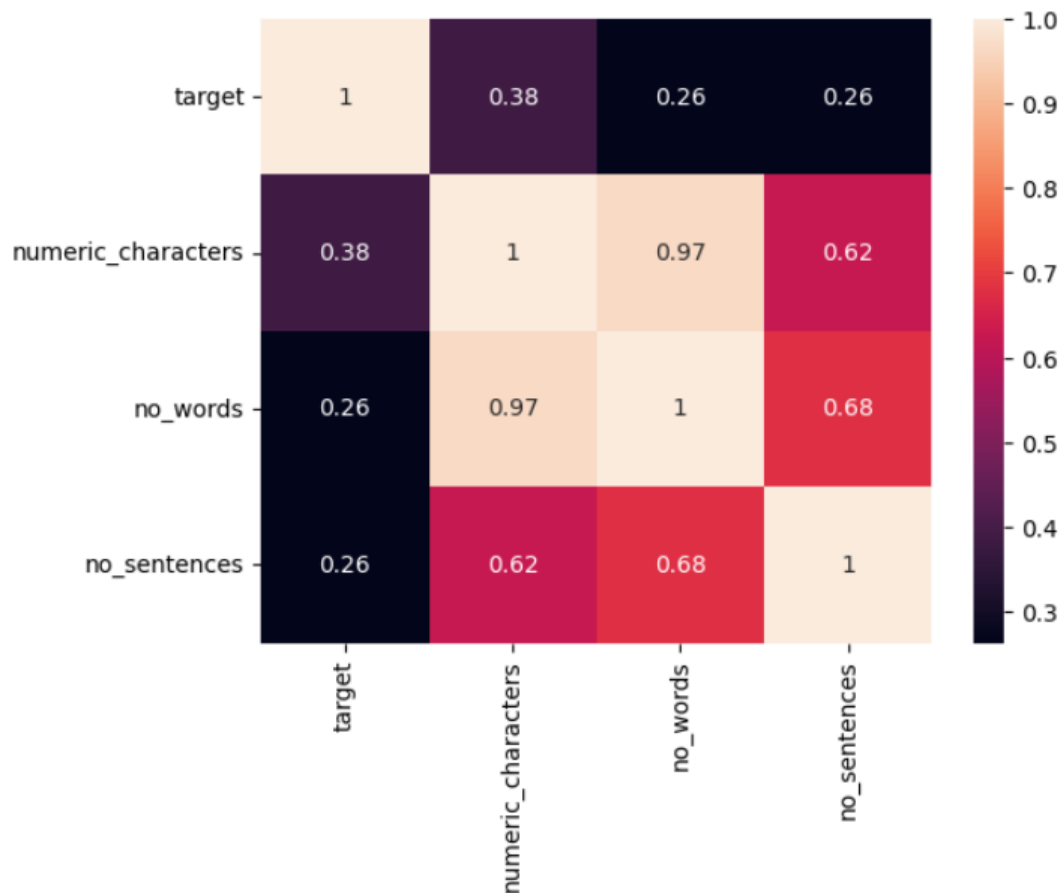
HeatMap :

Correlation (+Ve / -Ve Relationship Between Columns)

Numeric_Characters And No_Words Have A Very Strong Positive Correlation (0.97), Meaning Messages With More Words Almost Always Contain More Characters.

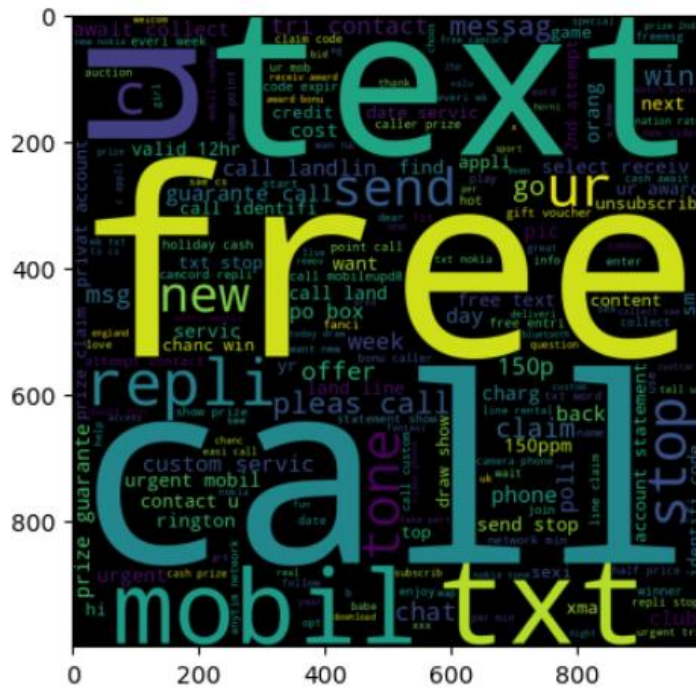
No_Sentences Has A Moderate Positive Correlation With Numeric_Characters (0.62) And No_Words (0.68), Indicating That Longer Messages Tend To Have More Sentences.

The Target Column Has A Weak Positive Correlation With All Three Features (0.26–0.38), Suggesting That Spam Messages Are Generally Longer, But These Features Alone Are Not Sufficient For Perfect Classification.



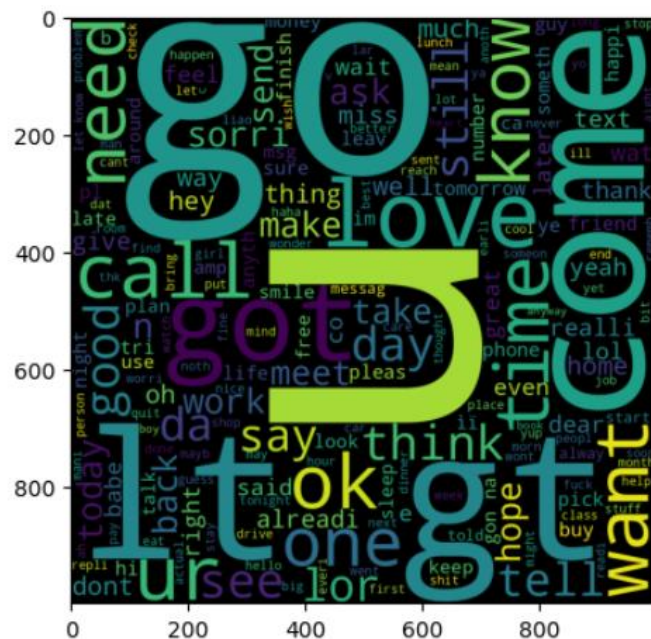
Word Cloud Of Spam Messages

Observe that Free, Call, Txt, Claim, Reply, And Win, Indicating That Spam Messages

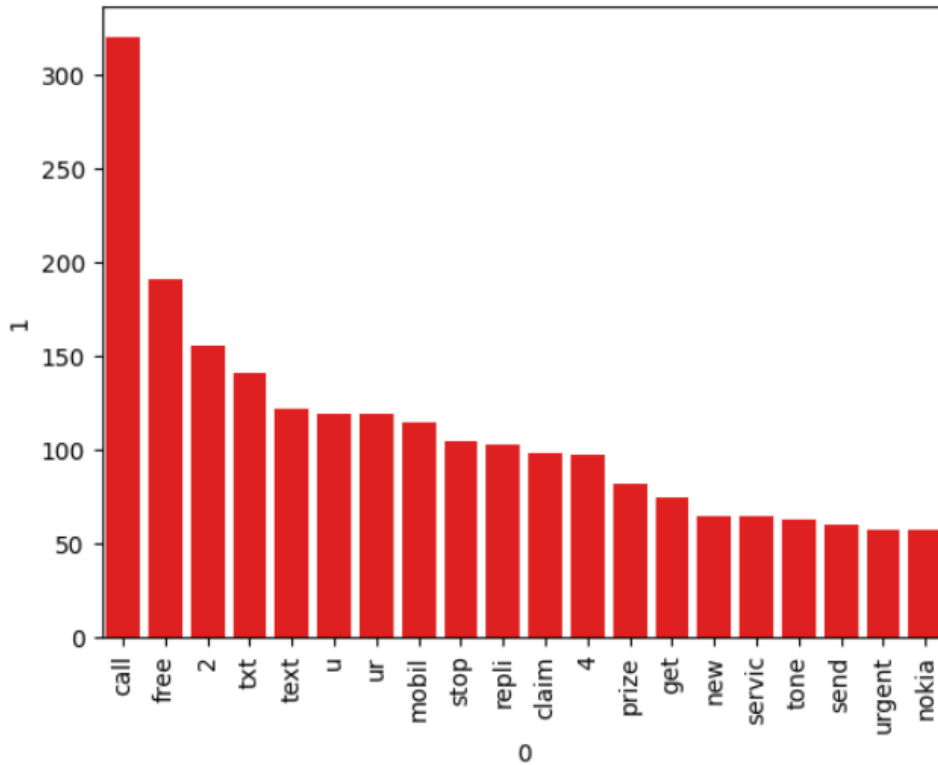


Word Cloud Of Not Spam Messages

Observe that Ok, Will, Call, Now, And Go, Indicating That Most Ham Messages



Bar Plot : Top 20 common words of spam messages



Bat Plot : Top 30 common words of ham messages

